



Approximation Algorithms for Big Data

Lecture 7: Kernel Density Estimation Cont'd

Spring 2026

Instructor: Tal Wagner

Administrative Announcements

- Class mid-point survey: [thanks everyone who filled it!](#)

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- Class mid-point survey: [thanks everyone who filled it!](#)
- Some things I'd like communicate back:
 - **There's only so much any lecture can do about technical understanding**
 - You basically have to engage with the technical details on your own (through HW etc.) to internalize them
 - **You have to communicate** to me things I don't otherwise have any way of knowing
 - Pace during lectures
 - Unclear points during lectures
 - Unfamiliar background material
 - Please remember I think of this class as a joint venture

Plan

- **Recap:** KDE approximation
 - Definitions
 - Sampling techniques
- **Today's topic:** Hashing-based estimators (HBE) for KDE
 - Background: importance sampling
 - Importance sampling for KDE via LSH
- **Final project information**

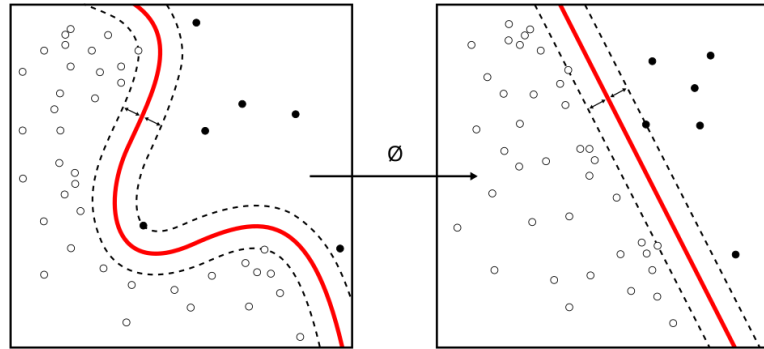
Recap: KDE Approximation

Kernels

- A **kernel** is a map $k: \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ that measures **similarity** between pairs of points
- Some commonly used kernels:
 - **Gaussian**: $k(x, y) = e^{-\|x-y\|_2^2/2\sigma^2}$
 - **Laplacian**: $k(x, y) = e^{-\|x-y\|_1/\sigma}$ or $k(x, y) = e^{-\|x-y\|_2/\sigma}$
 - **Polynomial** (of degree p): $k(x, y) = \left(\frac{x^T y}{\sigma^2} + c\right)^p$
- σ is the **bandwidth** parameter
 - Controls smoothness
 - We will often assume w.l.o.g. that $\sigma = 1$ by scaling the points (but not always)

Kernels as Inner Products

- Intuition: kernels are inner products in a transformed space
- Meaning, there is a map $\phi: \mathbb{R}^d \rightarrow \mathbb{R}^D$ such that $k(x, y) = \phi(x)^T \phi(y)$



- However, D is typically **very large or infinite** and thus not useful algorithmically
 - Note that in kernel SVM we never needed to compute ϕ or realize D

Kernel Approximation

- We saw two techniques:

- **Random features:** a distribution $\mathcal{F}$ is a distribution over functions $f: \mathbb{R}^d \rightarrow \mathbb{R}^m$ such that for all $x, y \in \mathbb{R}^d$:

$$k(x, y) = \mathbb{E}_{f \sim \mathcal{F}}[f(x)^T f(y)]$$

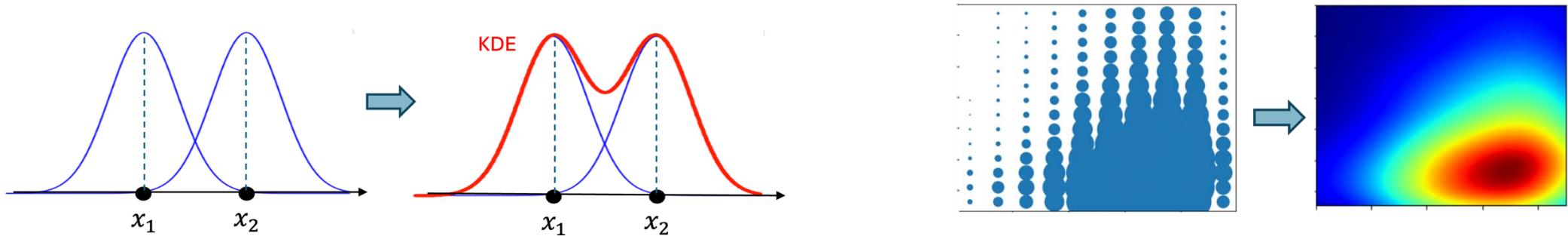
- **LSH:** a distribution $\mathcal{H}$ over hash functions h such that for all $x, y \in X$:

$$k(x, y) = \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]$$

- We saw **Random Fourier Features** for the Gaussian kernel
- We saw **LSH** for the Laplacian kernel

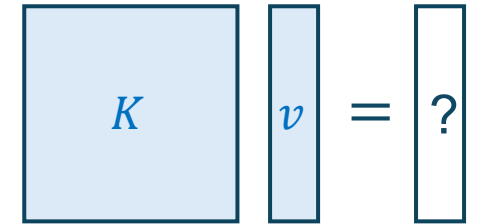
Kernel Density Estimation (KDE)

- Input: Set of n points $X \subset \mathbb{R}^d$
- Task: Given a query $y \in \mathbb{R}^d$, return $KDE_X(y) := \frac{1}{|X|} \sum_{x \in X} k(x, y)$
- Goal: Answer queries approximately with high probability



Extra Motivation for KDE: Kernel Matrices

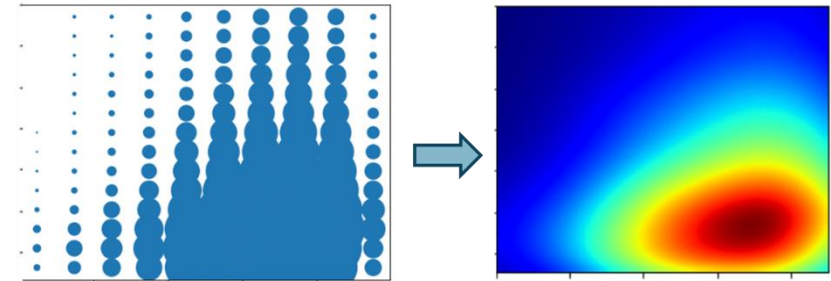
- Suppose we have a kernel $k(x, y)$ and points $x_1, \dots, x_n \in \mathbb{R}^d$
- Recall the **kernel matrix**: $K \in \mathbb{R}^{n \times n}$ defined as $K_{ij} = k(x_i, y_i)$
- Consider the kernel matrix-vector multiplication (KMVM) problem:
 - Given a vector $v \in \mathbb{R}^d$, return an approximation of Kv
 - **Exact computation time?**
 - Since K has **specific structure** (a kernel matrix), can we do better?


$$K v = ?$$

- **Observe:** $(Kv)_j = \sum_{i=1}^n K_{ij} v_i = \sum_{i=1}^n v_i \cdot k(x_i, y)$
- This is a weighted variant of KDE – approximation techniques for KDE are useful
- KMVM arises often in machine learning, large language models, etc. (we'll see more)

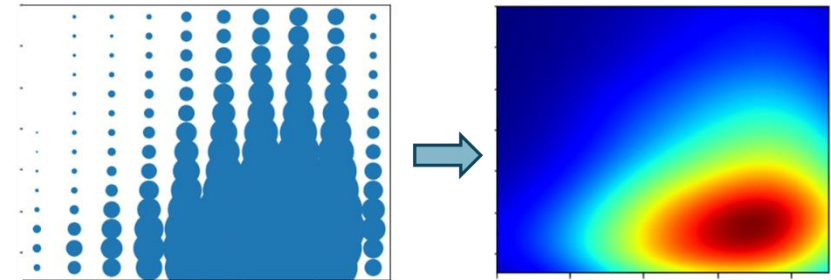
KDE Data Structures

- A KDE data structure has two operations:
 - Build: Given n points $X \subset \mathbb{R}^d$, build data structure
 - Query: Given $y \in \mathbb{R}^d$, return an approximation $\tilde{K}(y)$ of $KDE_X(y)$
 - Goal: The $\tilde{K}(y)$ needs to be a good approximation with high probability



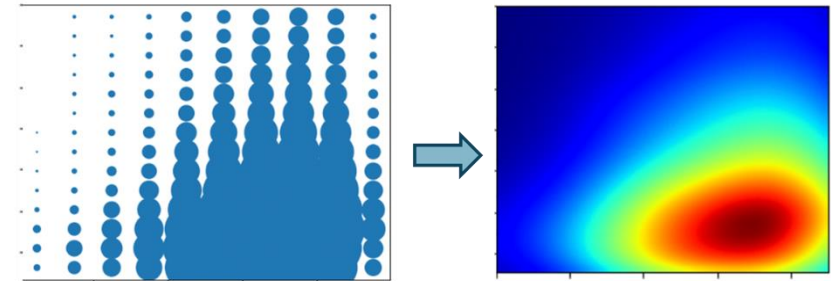
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 - Build time
 - Memory usage
 - Query time (main measure)



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 - Query time (main measure)
- Simplifying assumptions throughout:
 - $k(x, y)$ takes values in $[0, 1]$
 - $k(x, y)$ takes time $O(d)$ to compute



} Both holds for Gaussian, Laplacian, etc

KDE with Additive Error (1): Random Features

- Suppose we have random features: $k(x, y) = \mathbb{E}_{f \sim \mathcal{F}}[f(x)^T f(y)]$
- How can we get a KDE data structure?
 - Build:
 - Sample r random features $f_1, \dots, f_r \sim \mathcal{F}$
 - For every f_j , compute and store $F_j(X) := \frac{1}{|X|} \sum_{x \in X} f_j(x)$
 - Query: given y , return $\tilde{K}(y) = \frac{1}{r} \sum_{j=1}^r F_j(X)^T f_j(y)$

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- What are build time, memory, query time? For concreteness, analyze RFF
- Claim: if $\forall f, x, y$ we have $|f(x)^T f(y)| = O(1)$, and we take $r = O(\log(\delta^{-1})/\varepsilon^2)$,

$$\forall x, y, \quad \Pr[|KDE_X(y) - \tilde{K}(y)| \geq \varepsilon] \leq \delta$$

- Proof: Hoeffding

KDE with Additive Error (2): Random Sampling

- Consider the data structure:
 - Build: sample m points $z_1, \dots, z_r$ iid uniformly at random from X (with replacement)
 - Query: given y , return $\tilde{K}(y) = \frac{1}{r} \sum_{j=1}^r k(z_j, y)$

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- **Claim:** With sample size $r = O(\log(\delta^{-1})/\varepsilon^2)$, this is a KDE data structure with additive error ε and failure probability δ
 - Proof: HW2

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- Build time, memory, query time?
- **Claim:** With sample size $r = O(\log(\delta^{-1})/\varepsilon^2)$, this is a KDE data structure with additive error ε and failure probability δ
 - Proof: HW2
- Remark: This alludes to [coresets](#), a much broader topic in KDE and in general

KDE with Relative Error

- **Recall:** A KDE data structure has two operations:
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 - Query: Given $y \in \mathbb{R}^d$, return an approximation $\tilde{K}(y)$ of $KDE_X(y)$
- **Recall:** Additive error guarantee: Goal is $\tilde{K}(y) = KDE_X(y) \pm \varepsilon$ w.p. $1 - \delta$

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- **Now: Relative error guarantee**
 - Let $\varepsilon, \delta, \tau \in (0,1)$
 - Promise: the query y satisfies $KDE_X(y) \geq \tau$
 - Goal: $\tilde{K}(y) = (1 \pm \varepsilon) \cdot KDE_X(y)$ w.p. $1 - \delta$
- **Why is this better than additive error?** Consider $\tau \ll \varepsilon$

KDE with Relative Error via Additive Error

- KDE with **additive** error: $\tilde{K}(y) = KDE_X(y) \pm \varepsilon$ w.p. $1 - \delta$
- KDE with **relative** error: if $KDE_X(y) \geq \tau$, then $\tilde{K}(y) = (1 \pm \varepsilon) \cdot KDE_X(y)$ w.p. $1 - \delta$

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- KDE with **relative** error: if $KDE_X(y) \geq \tau$, then $\tilde{K}(y) = (1 \pm \varepsilon) \cdot KDE_X(y)$ w.p. $1 - \delta$
- If we do additive error $\tau\varepsilon$, then under the promise $KDE_X(y) \geq \tau$, we get:
 - $\tilde{K}(y) \leq KDE_X(y) + \tau\varepsilon \leq (1 + \varepsilon)KDE_X(y)$
 - $\tilde{K}(y) \geq KDE_X(y) - \tau\varepsilon \geq (1 - \varepsilon)KDE_X(y)$
 - *Relative error guarantee!*

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 - *Relative error guarantee!*
- What query time do we get from Random Features or Random Sampling?
 - $O\left(d \cdot \frac{\log(\delta^{-1})}{\tau^2 \varepsilon^2}\right)$

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 - *Relative error guarantee!*
- What query time do we get from Random Features or Random Sampling?
 - $O\left(d \cdot \frac{\log(\delta^{-1})}{\tau^2 \varepsilon^2}\right)$
 - This is a **suboptimal** analysis of Random Sampling – we'll see $O\left(d \cdot \frac{\log(\delta^{-1})}{\tau \varepsilon^2}\right)$

Recap: Median of Means

- Suppose we have a random variable Z with $\mathbb{E}[Z] = \mu$ and variance $\text{Var}(Z)$
 - For our current discussion, Z is an **unbiased KDE estimator**: $\mu = \text{KDE}_X(y)$
- **Theorem (MoM)**: with $r = O\left(\frac{\text{Var}(Z)}{\mu^2 \varepsilon^2} \log\left(\frac{1}{\delta}\right)\right)$ samples, we can get an estimate $\tilde{Z}$ that satisfies $\tilde{Z} = (1 \pm \varepsilon)\mu$ w.p. $1 - \delta$
 - Proof:
 - Let $\bar{Z}$ be the average of $\bar{r} = O\left(\frac{\text{Var}(Z)}{\mu^2 \varepsilon^2}\right)$ iid samples of Z
 - By Chebyshev's inequality, $\bar{Z} = (1 \pm \varepsilon)\mu$ w.p. $2/3$
 - Let $\tilde{Z}$ be the median of $O(\log(1/\delta))$ repetitions of $\bar{Z}$
 - By Chernoff's inequality, $\tilde{Z} = (1 \pm \varepsilon)\mu$ w.p. $1 - \delta$
- For simplicity, **we will suppress δ from now on**

KDE with Relative Error via Random Sampling

- As a baseline / warmup:
 - We'll prove that uniform random sampling gives KDE with relative error via MoM
 - We'll get that the number of samples needed is $O\left(\frac{\log(\delta^{-1})}{\tau \varepsilon^2}\right)$
 - This is better than the bound we get earlier for uniform random sampling through vanilla KDE relative error via additive error: $O\left(\frac{\log(\delta^{-1})}{\tau^2 \varepsilon^2}\right)$

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- Variance: $\text{Var}(Z) = \mathbb{E}[Z^2] - (\mathbb{E}[Z])^2 \leq \mathbb{E}[Z^2]$
 - $= \sum_{x \in X} \frac{1}{|X|} k(x, y)^2$
 - $\leq \max_{x^* \in X} k(x^*, y) \sum_{x \in X} \frac{1}{|X|} k(x, y)$
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- We got:

$$\mathbb{E}[Z] = \mu \text{ and } \text{Var}(Z) \leq \mu$$

- By MoM: $O\left(\frac{\text{Var}(Z)}{\mu^2 \varepsilon^2}\right) = O\left(\frac{1}{\mu \varepsilon^2}\right)$ samples suffice
- Since $\mu \geq \tau$, $O\left(\frac{1}{\tau \varepsilon^2}\right)$ samples suffice.



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- Remark: We don't need the median with Random Sampling, it's enough to take the average of all samples.

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- **Our next goal is to find better estimators for MoM via non-uniform sampling**

Importance Sampling for KDE

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- Goal: Improve the dependence on τ to $\left(\frac{1}{\tau^\alpha \varepsilon^2}\right)$ for some $\alpha \in (0,1)$
- How do we get better sample size, i.e., how can we reduce the variance?

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 - Let $x_1, \dots, x_n$ be the points in X
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 - Let $p_1, \dots, p_n \in [0,1]$ be corresponding probabilities such that $\sum_{i=1}^n p_i = 1$
 - **Importance sampling estimator:**
 - Sample x_i from X with probability p_i
 - Estimator: $Z := \frac{1}{n \cdot p_i} k(x_i, y)$

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???

First and Second Moments

- $\mathbb{E}[Z] = \sum_{i=1}^n p_i \cdot \frac{1}{n \cdot p_i} k(x_i, y) = KDE_X(y)$
- $\mathbb{E}[Z^2] = \sum_{i=1}^n p_i \cdot \left(\frac{1}{n \cdot p_i} k(x_i, y) \right)^2 = \frac{1}{n} \sum_{i=1}^n \frac{k(x_i, y)^2}{p_i}$

First and Second Moments

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- Computing the optimal sampling probabilities is as slow as computing exact KDE
 - Not useful algorithmically
- Can we compute approximate importance sampling probabilities?
 - We need an efficient approximate way to estimate $k(x_i, y)$
 - Do you know any?

Approximate Importance Sampling for KDE

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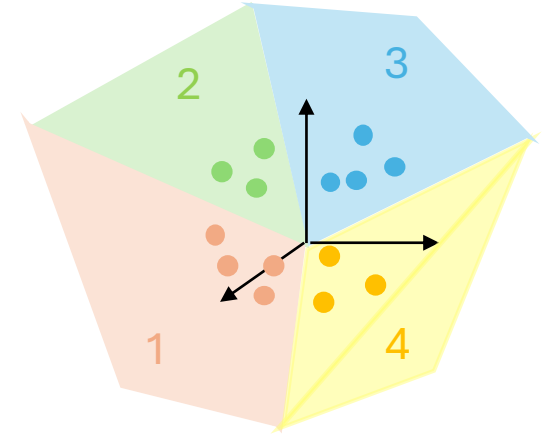
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- **Do you have ideas how?**

HBE: (Locality Sensitive) Hashing- Based Estimators for KDE

Charikar-Siminelakis '17

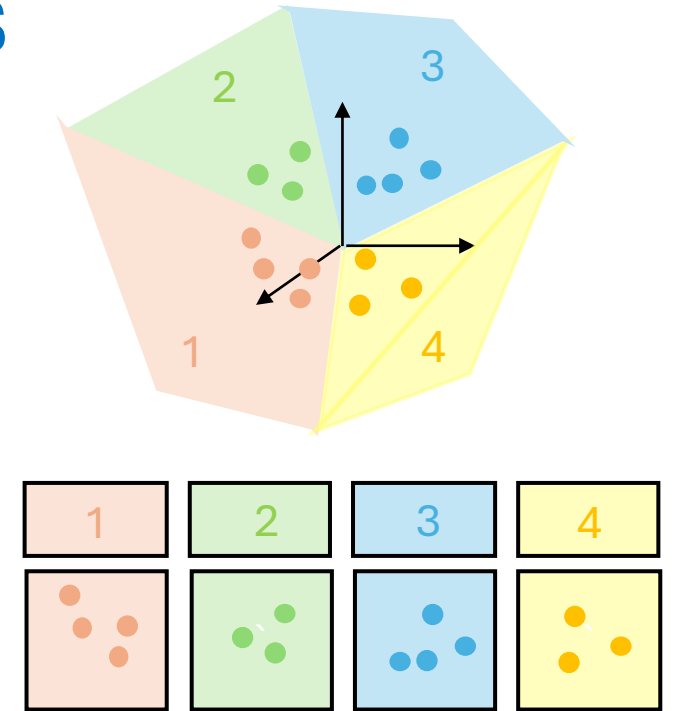
Recap: LSH for Nearest Neighbors

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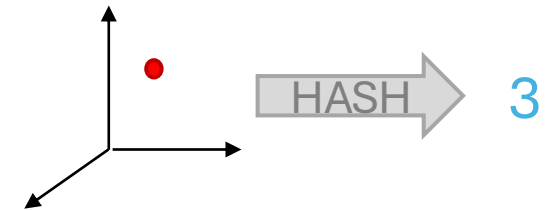
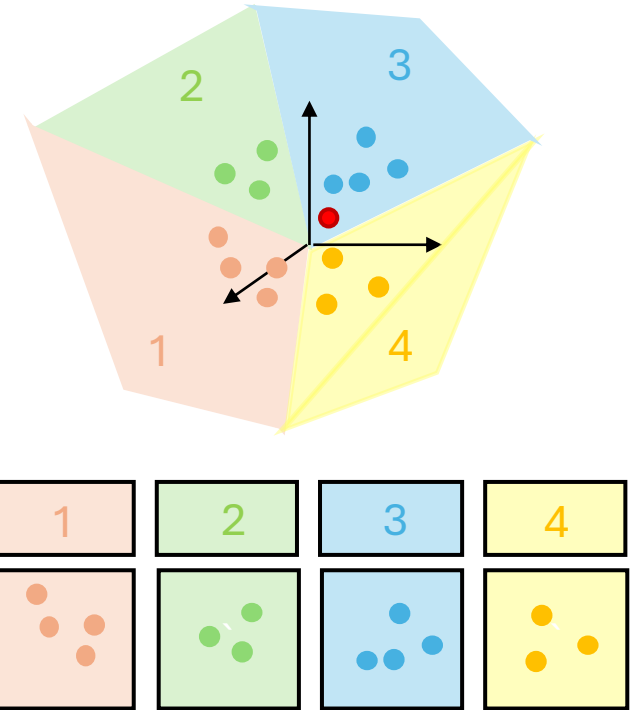
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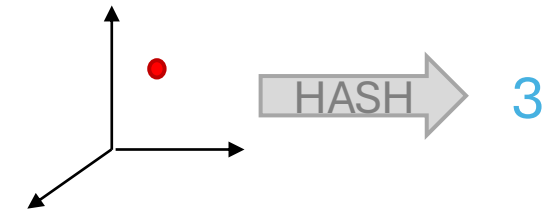
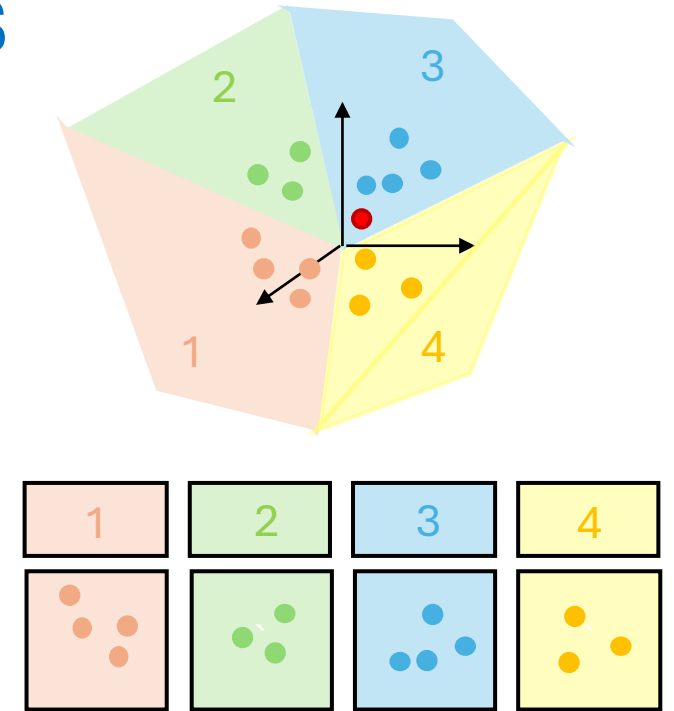
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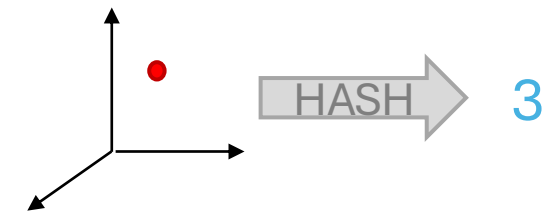
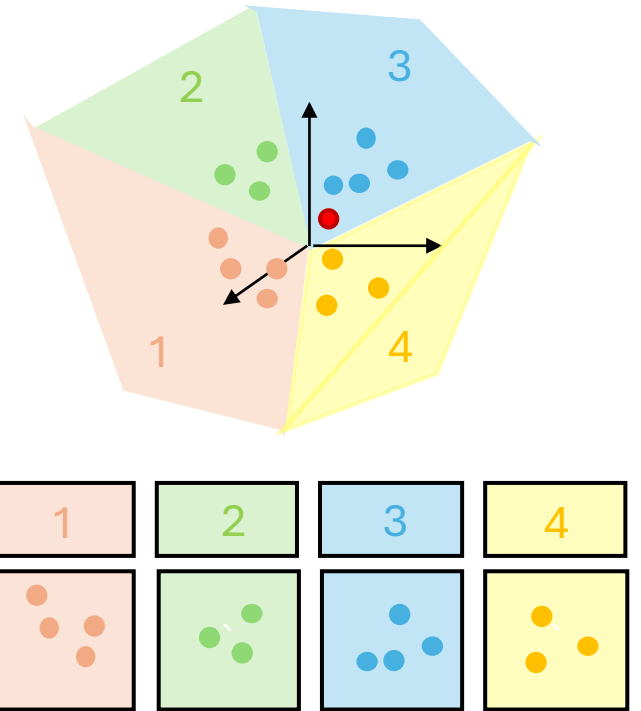
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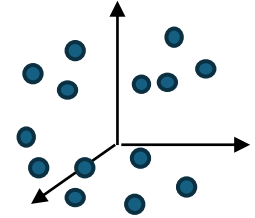
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- Can we use it for importance sampling?



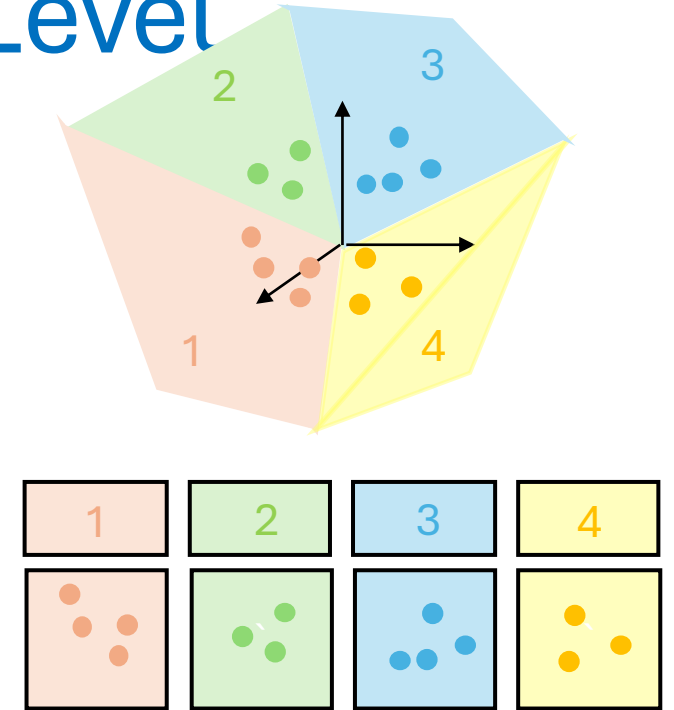
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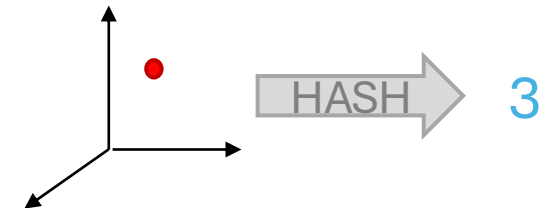
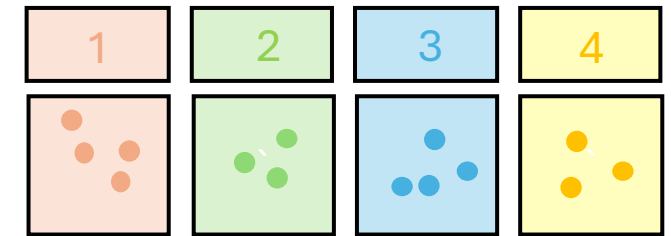
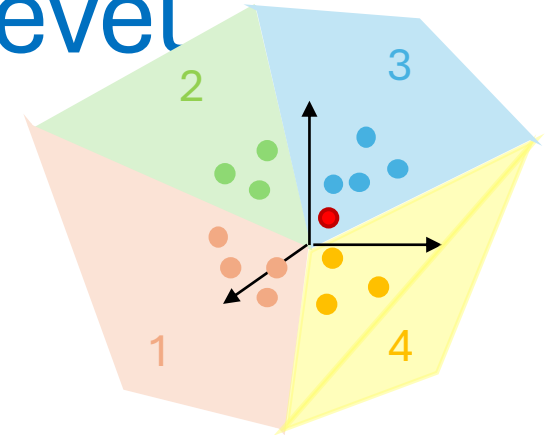
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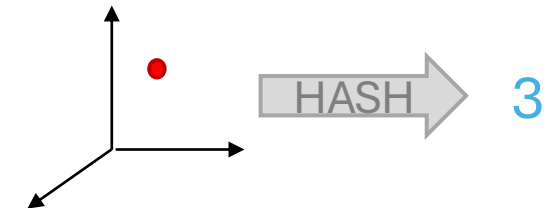
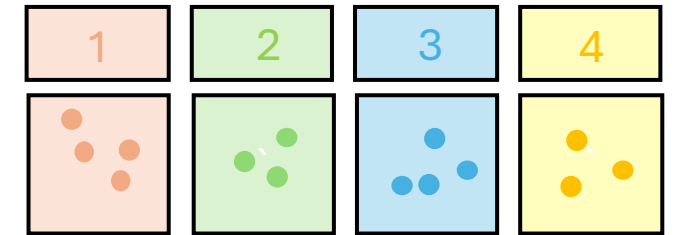
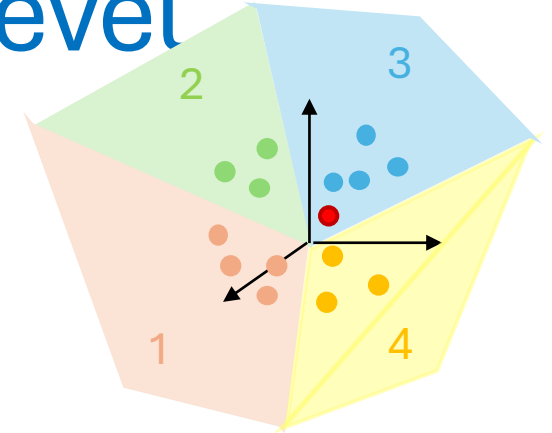
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- How does it compare to uniformly random sampling?



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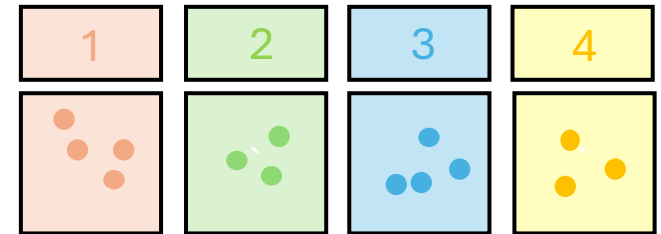
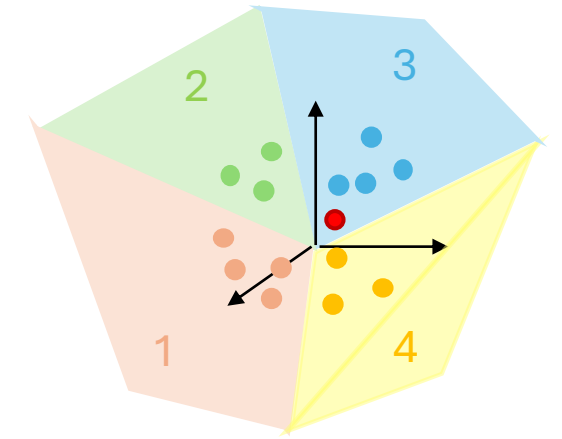
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 - For HBE we will use $\frac{1}{2}$ -LSHability (we will soon see why $\frac{1}{2}$):

$$\sqrt{k(x, y)} = \Pr_{h \sim \mathcal{H}} [h(x) = h(y)]$$

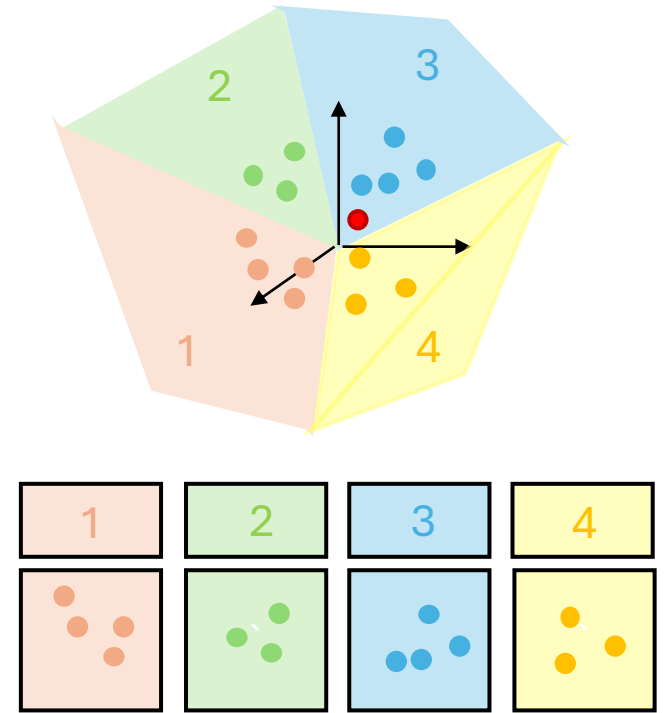
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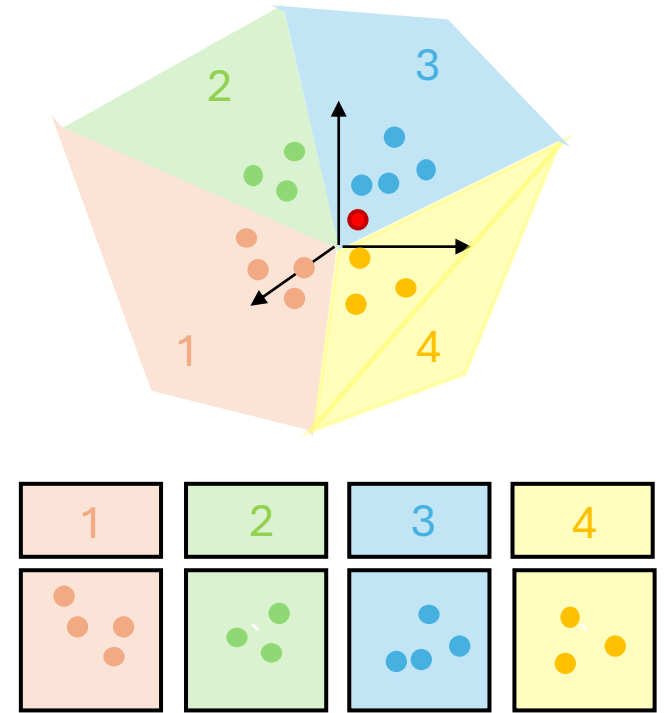


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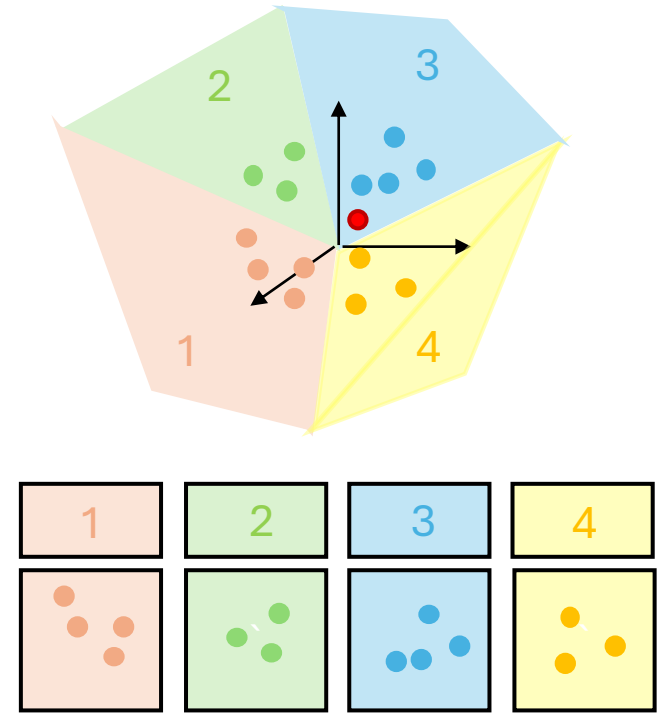


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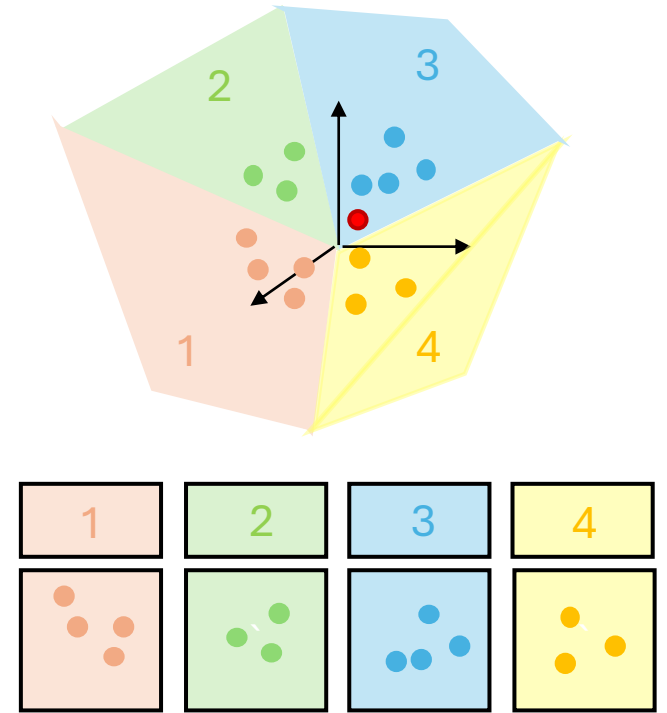
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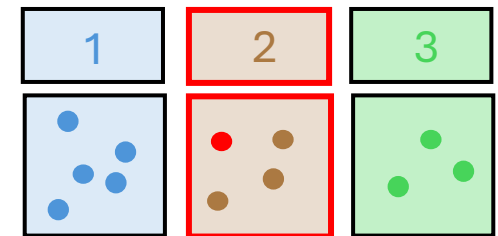
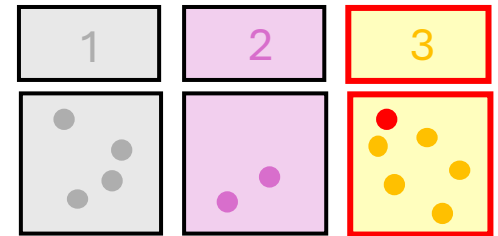
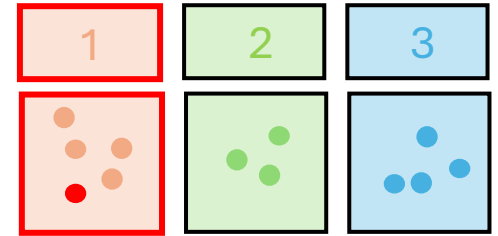
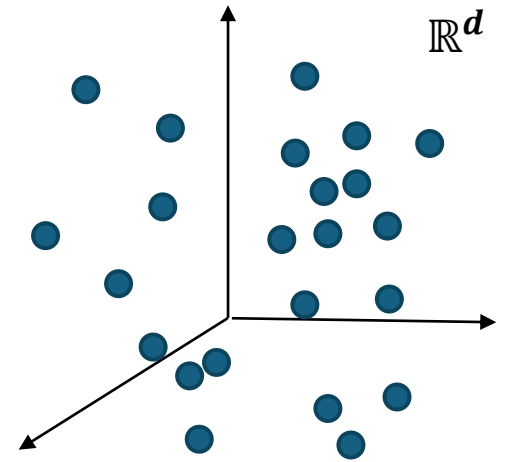
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HBE Data Structure

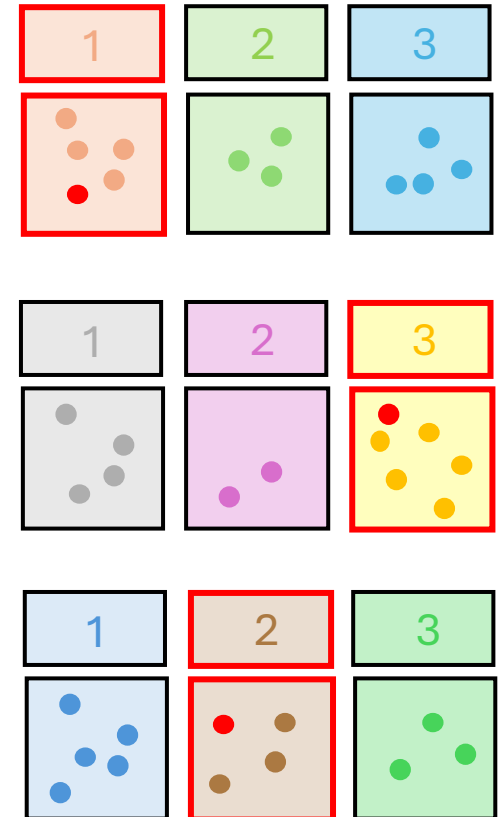
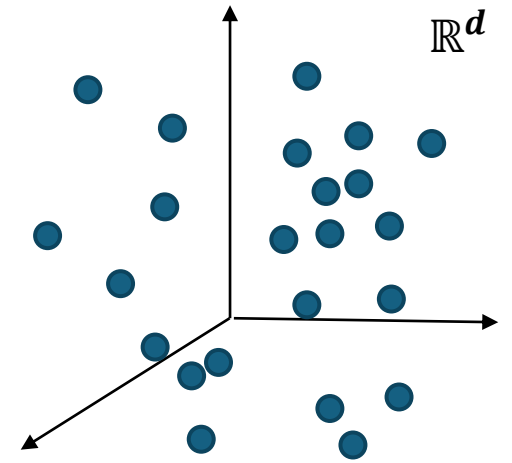
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- Our goal is to show that we can the desired guarantee:
 - $\Pr[\text{MoM}(Z_1, \dots, Z_r) = (1 \pm \varepsilon) \cdot KDE_X(y)] \geq 1 - \delta$
 - Provided that $KDE_X(y) \geq \tau$
- With much fewer repetitions than uniform random sampling:

$$r \ll O\left(\frac{\log(1/\delta)}{\varepsilon^2 \tau}\right)$$



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- $\mathbb{E}[Z] = \frac{1}{|X|} \mathbb{E}_{h^* \sim \mathcal{H}, x \sim B^*(y)} \left[\frac{|B^*(y)|}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y) \right]$
- $= \frac{1}{|X|} \mathbb{E}_{h^* \sim \mathcal{H}} \left[\mathbb{E}_{x \sim B^*(y)} \left[\frac{|B^*(y)|}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y) \mid h^* \right] \right]$
- $= \frac{1}{|X|} \mathbb{E}_{h^* \sim \mathcal{H}} \left[\sum_{x \in B^*(y)} \frac{1}{|B^*(y)|} \cdot \frac{|B^*(y)|}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y) \right]$
- $= \frac{1}{|X|} \mathbb{E}_{h^* \sim \mathcal{H}} \left[\sum_{x \in X} \mathbf{1}_{\{h^*(x) = h^*(y)\}} \cdot \frac{1}{|B^*(y)|} \cdot \frac{|B^*(y)|}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y) \right]$
- $= \frac{1}{|X|} \sum_{x \in X} k(x, y) \cdot \frac{1}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot \mathbb{E}_{h^* \sim \mathcal{H}} [\mathbf{1}_{\{h^*(x) = h^*(y)\}}]$
- $= \boxed{???$

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- $= \frac{1}{|X|} \mathbb{E}_{h^* \sim \mathcal{H}} \left[\sum_{x \in X} \mathbf{1}_{\{h^*(x) = h^*(y)\}} \cdot \frac{1}{|B^*(y)|} \cdot \frac{|B^*(y)|}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y) \right]$
- $= \frac{1}{|X|} \sum_{x \in X} k(x, y) \cdot \frac{1}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot \mathbb{E}_{h^* \sim \mathcal{H}} [\mathbf{1}_{\{h^*(x) = h^*(y)\}}]$
- $= \frac{1}{|X|} \sum_{x \in X} k(x, y) = KDE_X(y)$

HBE: Second Moment (1/4)

$$Z := \frac{|B^*(y)|}{|X| \cdot \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y)$$

HBE: Second Moment (1/4)

$$Z := \frac{|B^*(y)|}{|X| \cdot \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y)$$

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$$\bullet = \frac{1}{|X|^2} \mathbb{E}_{h^* \sim \mathcal{H}} \left[\mathbb{E}_{x \sim B^*(y)} \left[\left(\frac{|B^*(y)|}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y) \right)^2 \mid h^* \right] \right]$$

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$$\bullet = \frac{1}{|X|^2} \mathbb{E}_{h^* \sim \mathcal{H}} \left[\sum_{x \in X} \mathbf{1}_{\{h^*(x) = h^*(y)\}} \cdot \frac{1}{|B^*(y)|} \left(\frac{|B^*(y)|}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y) \right)^2 \right]$$

HBE: Second Moment (2/4)

$$Z := \frac{|B^*(y)|}{|X| \cdot \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y)$$

$$\bullet \frac{1}{|X|^2} \mathbb{E}_{h^* \sim \mathcal{H}} \left[\sum_{x \in X} \mathbf{1}_{\{h^*(x) = h^*(y)\}} \cdot \frac{1}{|B^*(y)|} \left(\frac{|B^*(y)|}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y) \right)^2 \right]$$

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- $\frac{1}{|X|^2} \mathbb{E}_{h^* \sim \mathcal{H}} \left[\sum_{x \in X} \mathbf{1}_{\{h^*(x) = h^*(y)\}} \cdot \frac{1}{|B^*(y)|} \left(\frac{|B^*(y)|}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y) \right)^2 \right]$
- $= \frac{1}{|X|^2} \mathbb{E}_{h^* \sim \mathcal{H}} \left[\sum_{x \in X} \mathbf{1}_{\{h^*(x) = h^*(y)\}} \sum_{z \in X} \mathbf{1}_{\{h^*(z) = h^*(y)\}} \left(\frac{k(x, y)}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \right)^2 \right]$

HBE: Second Moment (2/4)

$$Z := \frac{|B^*(y)|}{|X| \cdot \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y)$$

$$\begin{aligned} & \bullet \frac{1}{|X|^2} \mathbb{E}_{h^* \sim \mathcal{H}} \left[\sum_{x \in X} \mathbf{1}_{\{h^*(x) = h^*(y)\}} \cdot \frac{1}{|B^*(y)|} \left(\frac{|B^*(y)|}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y) \right)^2 \right] \\ & \bullet = \frac{1}{|X|^2} \mathbb{E}_{h^* \sim \mathcal{H}} \left[\sum_{x \in X} \mathbf{1}_{\{h^*(x) = h^*(y)\}} \sum_{z \in X} \mathbf{1}_{\{h^*(z) = h^*(y)\}} \left(\frac{k(x, y)}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \right)^2 \right] \\ & \bullet = \frac{1}{|X|^2} \sum_{x \in X} \left(\frac{k(x, y)}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \right)^2 \sum_{z \in X} \mathbb{E}_{h^* \sim \mathcal{H}} [\mathbf{1}_{\{h^*(x) = h^*(y)\}} \mathbf{1}_{\{h^*(z) = h^*(y)\}}] \end{aligned}$$

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 & \bullet = \frac{1}{|X|^2} \sum_{x \in X} \left(\frac{k(x, y)}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \right)^2 \sum_{z \in X} \mathbb{E}_{h^* \sim \mathcal{H}} [\mathbf{1}_{\{h^*(x) = h^*(y)\}} \mathbf{1}_{\{h^*(z) = h^*(y)\}}] \\
 & \bullet \leq \frac{1}{|X|^2} \sum_{x \in X} \left(\frac{k(x, y)}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \right)^2 \sum_{z \in X} \mathbb{E}_{h^* \sim \mathcal{H}} [\mathbf{1}_{\{h^*(z) = h^*(y)\}}]
 \end{aligned}$$

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 \end{aligned}$$

HBE: Second Moment (3/4)

$$Z := \frac{|B^*(y)|}{|X| \cdot \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y)$$

- Suppose that the kernel k is β -LSHable: $k(x, y)^\beta = \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]$.

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- Suppose that the kernel k is β -LSHable: $k(x, y)^\beta = \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]$.

- $\frac{1}{|X|^2} \sum_{x \in X} \left(\frac{k(x, y)}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \right)^2 \sum_{z \in X} \Pr_{h^* \sim \mathcal{H}}[h^*(z) = h^*(y)]$

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- $= \frac{1}{|X|^2} \sum_{x \in X} \left(\frac{k(x, y)}{k(x, y)^\beta} \right)^2 \sum_{z \in X} k(z, y)^\beta$

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$$\bullet = \frac{1}{|X|} \sum_{x \in X} k(x, y)^{2-2\beta} \cdot \frac{1}{|X|} \sum_{z \in X} k(z, y)^\beta$$

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- Suppose that the kernel k is β -LSHable: $k(x, y)^\beta = \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]$.
- $\frac{1}{|X|^2} \sum_{x \in X} \left(\frac{k(x, y)}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \right)^2 \sum_{z \in X} \Pr_{h^* \sim \mathcal{H}}[h^*(z) = h^*(y)]$
- $= \frac{1}{|X|^2} \sum_{x \in X} \left(\frac{k(x, y)}{k(x, y)^\beta} \right)^2 \sum_{z \in X} k(z, y)^\beta$
- $= \frac{1}{|X|} \sum_{x \in X} k(x, y)^{2-2\beta} \cdot \frac{1}{|X|} \sum_{z \in X} k(z, y)^\beta$
- **Claim:** Let $\alpha \in [0, 1]$ and $v_1, \dots, v_n \geq 0$. Then, $\frac{1}{n} \sum_{i=1}^n v_i^\alpha \leq \left(\frac{1}{n} \sum_{i=1}^n v_i \right)^\alpha$.

HBE: Second Moment (3/4) $Z := \frac{|B^*(y)|}{|X| \cdot \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y)$

- Suppose that the kernel k is β -LSHable: $k(x, y)^\beta = \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]$.

- $\frac{1}{|X|^2} \sum_{x \in X} \left(\frac{k(x, y)}{\Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \right)^2 \sum_{z \in X} \Pr_{h^* \sim \mathcal{H}}[h^*(z) = h^*(y)]$

- $= \frac{1}{|X|^2} \sum_{x \in X} \left(\frac{k(x, y)}{k(x, y)^\beta} \right)^2 \sum_{z \in X} k(z, y)^\beta$

- $= \frac{1}{|X|} \sum_{x \in X} k(x, y)^{2-2\beta} \cdot \frac{1}{|X|} \sum_{z \in X} k(z, y)^\beta$

- **Claim:** Let $\alpha \in [0, 1]$ and $v_1, \dots, v_n \geq 0$. Then, $\frac{1}{n} \sum_{i=1}^n v_i^\alpha \leq \left(\frac{1}{n} \sum_{i=1}^n v_i \right)^\alpha$.

- **Proof?**

HBE: Second Moment (4/4)

$$Z := \frac{|B^*(y)|}{|X| \cdot \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y)$$

- Claim: Let $\alpha \in [0,1]$ and $v_1, \dots, v_n \geq 0$. Then, $\frac{1}{n} \sum_{i=1}^n v_i^\alpha \leq \left(\frac{1}{n} \sum_{i=1}^n v_i \right)^\alpha$.

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$$Z := \frac{|B^*(y)|}{|X| \cdot \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y)$$

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- If $\beta \in [0.5,1]$ then we have both $2 - 2\beta \in [0,1]$ and $\beta \in [0,1]$, therefore:
- $\mathbb{E}[Z^2] \leq \frac{1}{|X|} \sum_{x \in X} k(x, y)^{2-2\beta} \cdot \frac{1}{|X|} \sum_{z \in X} k(z, y)^\beta$

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- $\leq \left(\frac{1}{|X|} \sum_{x \in X} k(x, y) \right)^{2-2\beta} \cdot \left(\frac{1}{|X|} \sum_{z \in X} k(z, y) \right)^\beta$

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- $\leq \left(\frac{1}{|X|} \sum_{x \in X} k(x, y) \right)^{2-2\beta} \cdot \left(\frac{1}{|X|} \sum_{z \in X} k(z, y) \right)^\beta$
- $\leq \left(\frac{1}{|X|} \sum_{z \in X} k(x, y) \right)^{2-\beta}$

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- $\leq \left(\frac{1}{|X|} \sum_{z \in X} k(x, y) \right)^{2-\beta}$
- $=$???

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- $\leq \left(\frac{1}{|X|} \sum_{x \in X} k(x, y) \right)^{2-2\beta} \cdot \left(\frac{1}{|X|} \sum_{z \in X} k(z, y) \right)^\beta$
- $\leq \left(\frac{1}{|X|} \sum_{z \in X} k(x, y) \right)^{2-\beta}$
- $= (KDE_X(y))^{2-\beta}$

HBE: Finishing Up

$$Z := \frac{|B^*(y)|}{|X| \cdot \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y)$$

- Recall we denote $\mu := KDE_X(y)$ for brevity

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$$Z := \frac{|B^*(y)|}{|X| \cdot \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y)$$

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- We have shown: $\mathbb{E}[Z] = \mu$ and $\text{Var}(Z) \leq \mathbb{E}[Z^2] = \mu^{2-\beta}$ as long as $\beta \in [0.5, 1]$

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- Recall: In MoM we need $r = O\left(\frac{\text{Var}(Z)}{\mu^2 \varepsilon^2}\right)$ repetitions to get our guarantee

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- We get: $r = O\left(\frac{1}{\mu^\beta \varepsilon^2}\right)$

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- Recall: In MoM we need $r = O\left(\frac{\text{Var}(Z)}{\mu^2 \varepsilon^2}\right)$ repetitions to get our guarantee
- We get: $r = O\left(\frac{1}{\mu^\beta \varepsilon^2}\right)$
- For the Laplacian kernel we can choose any $\beta \in (0, 1]$, so we choose $\beta = 0.5$

HBE: Finishing Up

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- Recall we denote $\mu := KDE_X(y)$ for brevity
- We have shown: $\mathbb{E}[Z] = \mu$ and $\text{Var}(Z) \leq \mathbb{E}[Z^2] = \mu^{2-\beta}$ as long as $\beta \in [0.5, 1]$
- Recall: In MoM we need $r = O\left(\frac{\text{Var}(Z)}{\mu^2 \varepsilon^2}\right)$ repetitions to get our guarantee
- We get: $r = O\left(\frac{1}{\mu^\beta \varepsilon^2}\right)$
- For the Laplacian kernel we can choose any $\beta \in (0, 1]$, so we choose $\beta = 0.5$
- Recalling we have the promise $\mu \geq \tau$ for every query y , we get:

- **Theorem:** For the Laplacian kernel, there is a KDE data structure with the relative approximation guarantee and running time $O\left(\frac{d}{\tau^{0.5} \varepsilon^2}\right)$. (why the d ?)

HBE: Finishing Up

$$Z := \frac{|B^*(y)|}{|X| \cdot \Pr_{h \sim \mathcal{H}}[h(x) = h(y)]} \cdot k(x, y)$$

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- This is asymptotically better than uniform random sampling which required $O\left(\frac{d}{\tau \varepsilon^2}\right)$

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- By now there are better bounds for relative-error KDE through LSH beyond HBE
 - We may see a high-level overview of the method
 - Can you think of a way?

Concluding Remarks: Current KDE Bounds

- [Charikar-Kapralov-Nouri-Siminlakis '20]:
 - Their simpler method gets $\approx 1/\tau^{0.25}\epsilon^2$ for both Gaussian and Laplacian
 - Their more complicated method gets $\approx 1/\tau^{0.173}\epsilon^2$ for Gaussian and $\approx 1/\tau^{0.1}\epsilon^2$ for the ℓ_2 -Laplacian kernel $e^{-\|x-y\|_2}$
- Optimal bounds (for, say, the Gaussian kernel) still open and research is ongoing
- Possible project topic (either deep reading or research)

Questions?

Next: Final Project Information

Final Project Information

- Reminder:
 - Choose either **deep reading** or **research** project
 - **Deep reading:**
 - Choose 1-3 papers related to class
 - Write a project that adds something over them
 - E.g., better or different presentation; unified exposition of a few papers; etc
 - **Research:** Try to make progress on an open problem related to class
 - Can fall back from research to deep reading at any time
- Grade will be based on **written final report** and on **presentation to instructor**

Final Project Information: Timeline

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 - Chosen paper(s) and a short description of what you intend to do
 - Choose papers from lists on Moodle or other ones with my approval
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- **~ Aug 9 –31:** Do your [project presentation](#) after I have had time to read them
 - I will publish time slots
 - Will likely be over zoom
 - If you need special scheduling accommodations, **work with me in advance**

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 - No expository “high level” reports lacking mathematical depth
- Choose or formulate **key result to focus on** (the report will have a page limit)
 - Goal: Present a result that conveys the **key algorithmic and analytic ideas** of the paper – “separate the wheat from the chaff”
 - **Strike a balance**: don’t over-simplify; don’t get mired in unnecessary details
 - Part of the challenge you are evaluated on is **distilling these main idea**

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- **Questions?**